STATISTICS 652

LECTURE 14 - 2026

COMMENT: ASYMPTOTIC EQUIVALENCE BETWEEN LRT AND SCORES/WALDS TEST

REFERENCE TO REGULARITY CONDITIONS FOR LARGE SAMPLE THEORY FOR M.L.E.

- Rao, C.R. : CHAPTER 5, SECTION f. CHAPTER 6, SECTION e.

RECALL WALD'S TYPE $\chi^2$-TEST AND CONFIDENCE REGION

GIVEN $\sqrt{n}(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}) \to_d N_q(\underset{\sim}{0}, \mathcal{I}_\theta)$ with $\mathcal{I}_\theta > 0$

and $\hat{\mathcal{I}}_\theta \to_p \mathcal{I}_\theta$ for all $\underset{\sim}{\theta} \in (H) \subset \mathbb{R}^q$

THEN $n(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta})' \hat{\mathcal{I}}_\theta^{-1} (\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}) \to_d \chi^2_q$

$\Rightarrow (1-\alpha)\times 100\%$ ASYMPTOTIC CONFIDENCE ELLIPSOID FOR $\underset{\sim}{\theta}$

$$C(\underset{\sim}{\theta}) = \left\{ \underset{\sim}{\theta} \in \mathbb{R}^q \;\middle|\; (\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta})' \hat{\mathcal{I}}_\theta^{-1} (\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}) \le \chi^2_{q,\alpha}/n \right\}$$

$$= \bigcap_{\substack{\underset{\sim}{a} \in \mathbb{R}^q \\ \underset{\sim}{a} \neq 0}} \left\{ \underset{\sim}{\theta} \in \mathbb{R}^q \;\middle|\; \underset{\sim}{a}'\underset{\sim}{\theta} \in \underset{\sim}{a}'\hat{\underset{\sim}{\theta}} \pm \sqrt{\chi^2_{q,\alpha}} \sqrt{\underset{\sim}{a}' \mathcal{I}_\theta^{-1} \underset{\sim}{a}} \right\}$$

by CLASS LEMMA 8.1.1 (OR HW PROBLEM 3.11)

SCHEFFE TYPE SIMULTANEOUS $(1-\alpha)\times 100\%$ ASYMPTOTIC CONFIDENCE INTERVAL over all $\underset{\sim}{a} \in \mathbb{R}^q$, $\underset{\sim}{a} \neq \underset{\sim}{0}$

$$(14.1) \qquad \underset{\sim}{a}'\hat{\underset{\sim}{\theta}} - \sqrt{\chi^2_{q,\alpha}} \sqrt{\underset{\sim}{a}' \hat{\mathcal{I}}_\theta^{-1} \underset{\sim}{a}} \le \underset{\sim}{a}'\underset{\sim}{\theta} \le \underset{\sim}{a}'\hat{\underset{\sim}{\theta}} + \sqrt{\chi^2_{q,\alpha}} \sqrt{\underset{\sim}{a}' \mathcal{I}_\theta^{-1} \underset{\sim}{a}}$$

THAT IS, PROBABILITY (14.1) HOLDS FOR EVERY $\underset{\sim}{a} \in \mathbb{R}^q$, $\underset{\sim}{a} \neq \underset{\sim}{0}$ GOES TO $(1-\alpha)$ as $n \to \infty$.

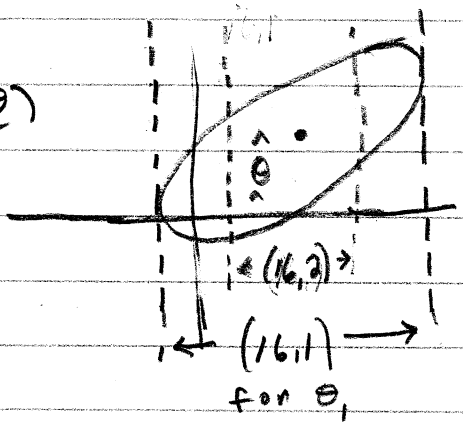
INDIVIDUAL $(1-\alpha)\times 100\%$ ASYMPOTIC CONFIDENCE INTERVAL FOR $\underset{\sim}{a}'\underset{\sim}{\theta}$

$$(14.2) \qquad \underset{\sim}{a}'\hat{\underset{\sim}{\theta}} - y_{\frac{\alpha}{2}} \sqrt{\underset{\sim}{a}' \underset{\theta}{\mathcal{I}}^{-1} \underset{\sim}{a}} \leq \underset{\sim}{a}'\underset{\sim}{\theta} \leq \underset{\sim}{a}'\hat{\underset{\sim}{\theta}} + y_{\frac{\alpha}{2}} \sqrt{\underset{\sim}{a}' \hat{\underset{\theta}{\mathcal{I}}} \underset{\sim}{a}}$$

SINCE $\sqrt{n}\left(\underset{\sim}{a}'\hat{\underset{\sim}{\theta}} - \underset{\sim}{a}'\underset{\sim}{\theta}\right) = \underset{\sim}{a}'\sqrt{n}\left(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}\right) \xrightarrow{d} N\left(0, \underset{\sim}{a}' \underset{\theta}{\mathcal{I}} \underset{\sim}{a}\right)$

and $\underset{\sim}{a}' \hat{\underset{\theta}{\mathcal{I}}} \underset{\sim}{a} \xrightarrow{p} \underset{\sim}{a}' \underset{\theta}{\mathcal{I}} \underset{\sim}{a}$.

HOWEVER; PROBABILITY (14.2) HOLDS FOR TWO DISTINCT $\underset{\sim}{a}_1 \in \mathbb{R}^q$ on $\underset{\sim}{a}_2 \in \mathbb{R}^q$ IS LESS THAN $(1-\alpha)$ (asymptotically),




NOTE: $y_{\frac{\alpha}{2}} < \sqrt{\chi^2_{q,\alpha}}$ $\quad q > 1$

OR $\chi^2_{1,\alpha} < \chi^2_{q,\alpha}$

FOR $H_0: \underset{\sim}{\theta} = \underset{\sim}{\theta}_0$ VS. $H_1: \underset{\sim}{\theta} \neq \underset{\sim}{\theta}_0$

WALD'S TYPE ASYMPOTOTIC $\alpha$-LEVEL CRITICAL REGION

$$(14.3) \qquad T_n = n\left(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}_0\right)^T \hat{\underset{\theta}{\mathcal{I}}}^{-1} \left(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}_0\right) > \chi^2_{q,\alpha}$$

and $T_n = \sup_{\substack{\underset{\sim}{a} \in \mathbb{R}^2 \\ \underset{\sim}{a} \neq 0}} Z_{\underset{\sim}{a}}^2$ where $Z_{\underset{\sim}{a}} = \dfrac{\underset{\sim}{a}'\hat{\underset{\sim}{\theta}} - \underset{\sim}{a}'\underset{\sim}{\theta}_0}{\sqrt{\underset{\sim}{a}' \hat{\Sigma}_0 \underset{\sim}{a}}}$

by CLASS LEMMA 7.6 (OR HW PROBLEM 3.10)

SCHEFFE TYPE SIMULTANEOUS $\alpha$-LEVEL ASYMPTOTIC REJECTION REGION FOR $H_{0,\underset{\sim}{a}}: \underset{\sim}{a}'\underset{\sim}{\theta} = \underset{\sim}{a}'\underset{\sim}{\theta}_0$ VS. $H_{1,\underset{\sim}{a}}: \underset{\sim}{a}'\underset{\sim}{\theta} \neq \underset{\sim}{a}'\underset{\sim}{\theta}_0$

(14.4) $\quad Z_{\underset{\sim}{a}}^2 > \chi^2_{1,\alpha}$ OR $|Z_{\underset{\sim}{a}}| > \sqrt{\chi^2_{1,\alpha}}$

THAT IS, PROBABILITY OF MAKING ONE TYPE I ERROR FOR $H_{0,\underset{\sim}{a}}$ over all $\underset{\sim}{a} \in \mathbb{R}^2$, $\underset{\sim}{a} \neq \underset{\sim}{0}$ GOES TO $\alpha$ AS $n \to \infty$.

SIMULTANEOUS TEST: $H_0 = \bigcap_{\substack{\underset{\sim}{a} \in \mathbb{R}^2 \\ \underset{\sim}{a} \neq \underset{\sim}{0}}} H_{0,\underset{\sim}{a}}$, $H_1 = \bigcup_{\substack{\underset{\sim}{a} \in \mathbb{R}^2 \\ \underset{\sim}{a} \neq \underset{\sim}{0}}} H_{1,\underset{\sim}{a}}$

REJECT $H_0$ USING (14.3)

$\Leftrightarrow$ REJECT $H_{0,\underset{\sim}{a}}$ USING (14.4) FOR SOME $\underset{\sim}{a} \in \mathbb{R}^2$, $\underset{\sim}{a} \neq \underset{\sim}{0}$

DO NOT REJECT $H_0$ USING (14.3)

$\Leftrightarrow$ DO NOT REJECT $H_{0,\underset{\sim}{a}}$ USING (14.4) FOR all $\underset{\sim}{a} \in \mathbb{R}^2$, $\underset{\sim}{a} \neq \underset{\sim}{0}$.

INDIVIDUAL ASYMPTOTIC $\alpha$-LEVEL CRITICAL REGION FOR $H_{0,\underset{\sim}{a}}$ VS $H_{1,\underset{\sim}{a}}$

(14.5) $\quad Z_{\underset{\sim}{a}}^2 > \chi^2_{1,\alpha}$ OR $|Z_{\underset{\sim}{a}}| > z_{\alpha/2}$.

NOTE: HAS APPLICATIONS TO MANY STATISTICAL METHODS

# BONFERONNI SIMULTANEOUS INFERENCE FOR TESTS AND CONF. REGIONS

- OVERALL TYPE I ERROR = EXPERIMENTWISE TYPE I ERROR
  PROBABILITY OF MAKING AT LEAST ONE TYPE I ERROR WHEN PERFORMING MULTIPLE TESTS (ASSUMING ALL NULL HYPOTHESES ARE TRUE)

- OVERALL CONFIDENCE LEVEL = EXPERIMENTWISE CONFIDENCE LEVEL
  PROBABILTY ALL CONFIDENCE REGIONS COVER THE CORRESPONDING PARAMETERS WHEN CONSTRUCTING MULTIPLE CONFIDENCE REGIONS

BONFERRONI'S METHOD (BASED ON SUBADDITIVITY)

- $P(A_1 \cup A_2 \cup \cdots \cup A_m) \le P(A_1) + \cdots + P(A_m) = m\alpha_0$ when $P(A_j) = \alpha_0$.
- SUPPOSE $A_j$ IS THE REJECTION REGION FOR $H_{0,j}$, $j = 1, \ldots, m$ with $P(A_j \mid H_{0,j}) = \alpha_0$. SO, $P(A_1 \cup \cdots \cup A_m \mid H_{0,1} \cap \cdots \cap H_{0,m}) \le m\alpha_0$

FOR SIMULTANEOUS TESTS: TO OBTAIN AN EXPERIMENTWISE TYPE I ERROR $\le \alpha_E$ CHOOSE $\alpha_0 = \alpha_E / m$ FOR EACH OF $m$ tests.

To guarantee an overall CONFIDENCE OF AT LEAST $(1-\alpha_E)\times 100\%$ WHEN constructing $m$ confidence regions, CHOOSE EACH REGION AT CONFIDENCE LEVEL TO BE AT LEAST $(1-\alpha_E/m)\times 100\%$

THAT IS, REPLACE $\sqrt{\chi^2_{2,\alpha}}$ with $t_{n-g,\frac{\alpha_E}{2m}}$ in (14.1)) and (14.4)

| BONFERONNI | vs. | SCHEFFE |
|---|---|---|
| ◦ VERY GENERAL | | ◦ SPECIFIC TO PROBLEM |
| ◦ m tests OR confidence regions MUST BE DECIDED WITHOUT LOOKING AT DATA: A-PRIORI | | ◦ TESTS AND CONFIDENCE REGIONS CAN BE DATA DEPENDENT; POST-PRIORI |
| ◦ AS m INCREASES ⇒ REDO ALL PREVIOUS TESTS AND CONF. REGIONS | | ◦ m INCREASES ⇒ NO CHANGE TO PREVIOUS TESTS AND CONF. REGIONS |
| ◦ BEST FOR SMALL m | | ◦ BETTER FOR LARGE m |

WHICH METHOD TO CHOOSE FOR A GIVEN m ? COMPARE

◦ $z_{\frac{\alpha_E}{2m}}$ TO ◦ $\sqrt{\chi^2_{q,\alpha_E}}$ CHOOS SMALLEST

EXAMPLE : m

| m | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $z_{\frac{.05}{2m}}$ | 1.96 | 2.24 | 2.39 | 2.50 | 2.58 | 2.63 | 2.69 | 2.73 | 2.77 | 2.81 |

| q | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $\sqrt{\chi^2_{q,.05}}$ | 1.96 | 2.45 | 2.80 | 3.08 |

FOR $q = 10$, BONFERONNI IS BETTER UNTIL $m = 2,658$

BOTH $\approx 4.278672$